

Module

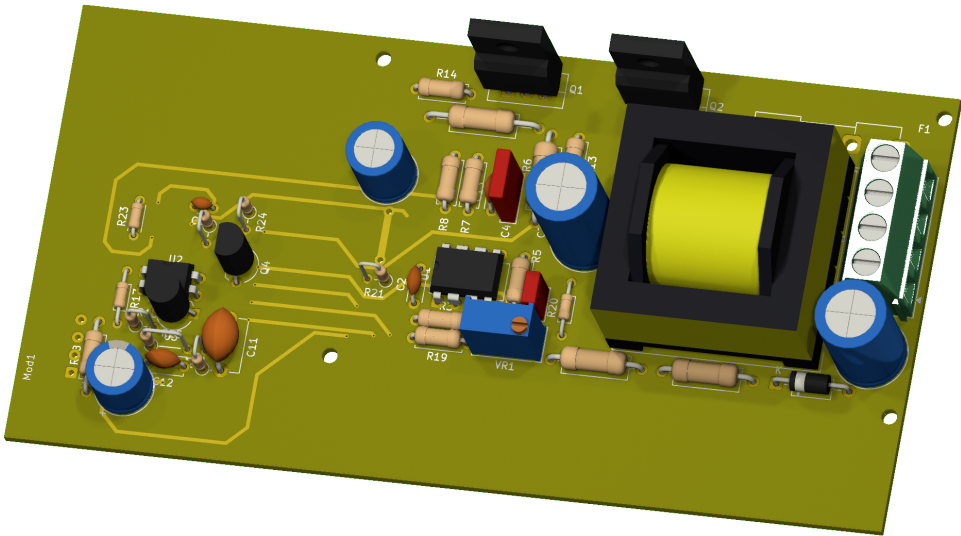
Variant: CHECKED

Rev + (Unreleased)

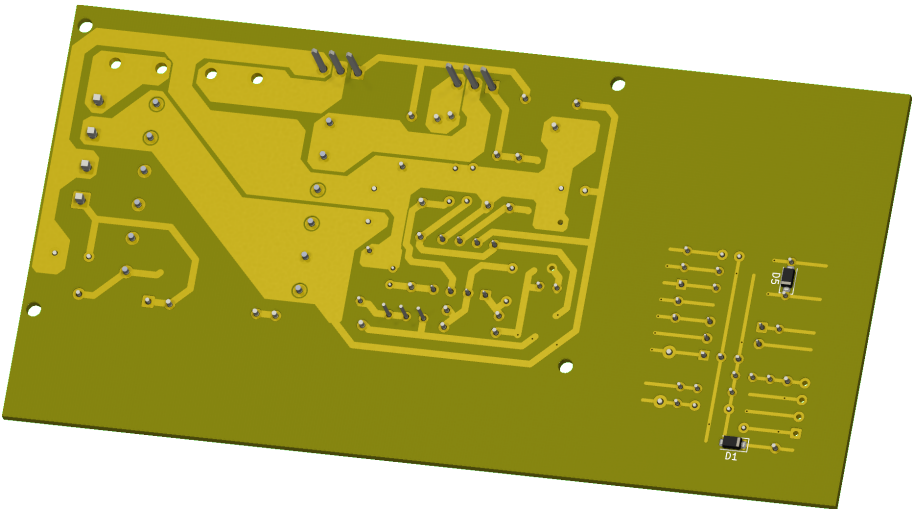
2026-01-24

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TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

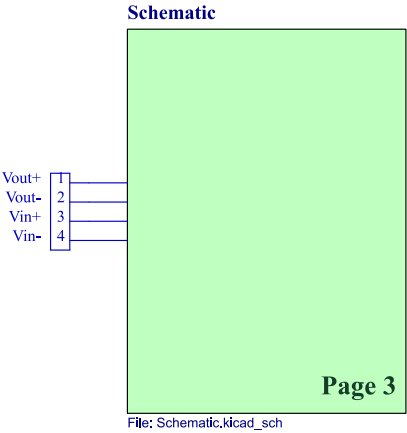
Example text for critical layout guidelines.

Date: 24-Jan-2026

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 120601b
	Sheet Title:	Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Path: /	File Name: Isolated_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: + (Unreleased)
		Reviewer: NA	Size: A3	Sheet: 1 of 4	

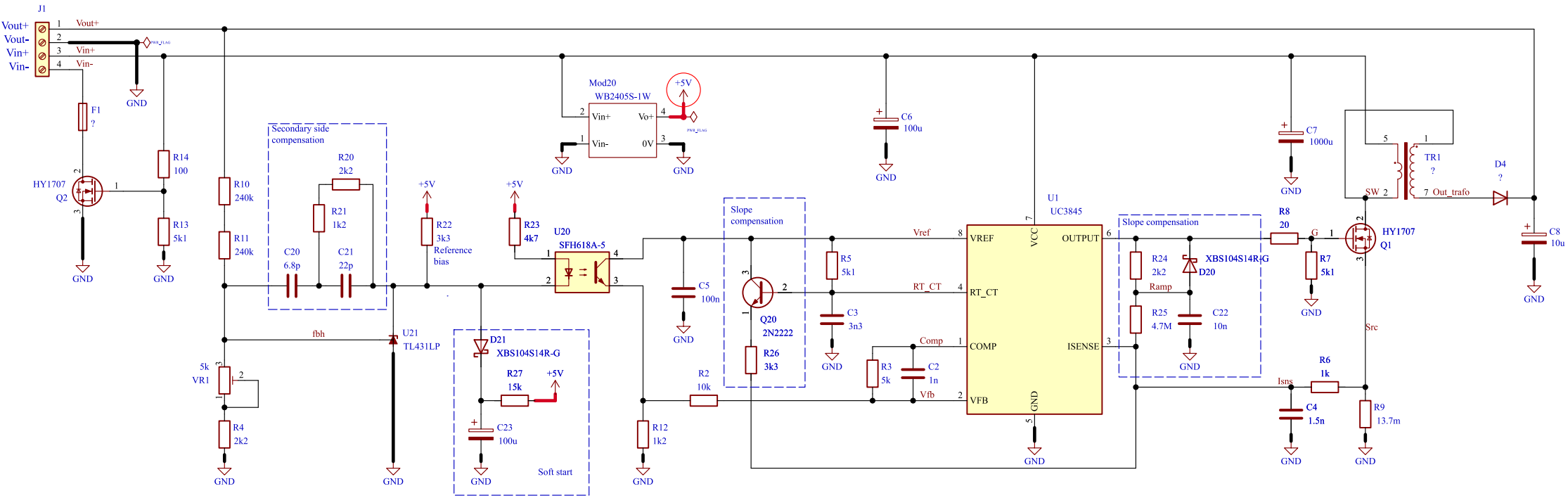
[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1



	Comments: AliExpress module example (link)		Company: FR		Variant: CHECKED		Git Hash: 120601b			
			Board Name: Module			Project Name: Isolated step-up 12:450 V				
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-12-20		Revision: + (Unreleased)	
	Sheet Path: /Block Diagram/				Reviewer: NA		Size: A4		Sheet: 2 of 4	

[3] Schematic



Estimated specs for unidentified / custom parts:
- F1: ~5 A
- D4: ~500 mA x 600 V
- R9: ~13.7 mΩ (110 mV @ 8.04 A)
- TR1: ~24:1 turns ratio, 10 μH : 5.76 mH

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 120601b
		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Schematic	File Name: Schematic.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: + (Unreleased)
	Sheet Path: /Block Diagram/Schematic/		Reviewer: NA	Size: A3	Sheet: 3 of 4

	1	2	3	4	5	6			
A	[4] Revision History					A			
B	Version 1.0.0 not found. Version 1.0.0 not found.	Version 1.0.1 - 2025-12-21				B			
C						C			
D						D			
			Comments:	Company:	Variant:	Git Hash:			
			Revision History	Board Name:	Project Name:				
				File Name:	Designer:	Date:	Revision:		
				Reviewer:	Size:	Sheet:			
			Sheet Path:						
	1	2	3	4	5	6			